

Enhancing Mobile Visualisation Interactivity: Insights on a Mixed-fidelity Prototyping Approach

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ABSTRACT

The dynamic nature of mobile devices presents significant challenges for mobile information visualisation design. The limited screen space of mobile devices requires increased user interaction to effectively accommodate information, posing challenges for touch interaction design. In turn, gesture design must balance ergonomics, adaptability to diverse contexts, and intuitive user experiences. Early design evaluation through low-fidelity prototypes is essential for detecting usability issues in a cost-effective and timely manner. To compensate the limited support for touch gestures and animated transitions offered by existing prototyping tools, we are developing a proof-of-concept for a tool that facilitates the creation of interactive animated prototypes of mobile visualisations. This paper presents the results of an exploratory test conducted on a preliminary version of our tool. We report four reflections about the challenges emerged during testing that, we believe, can foster further discussions and research towards obtaining tools specifically tailored to mobile infovis prototyping.

CCS CONCEPTS

• **Human-centered computing** → **Systems and tools for interaction design**; Visualization design and evaluation methods; *Information visualization*.

KEYWORDS

Prototyping, Interaction Design, Mobile Information Visualisation, Exploratory Test

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1 INTRODUCTION

Mobile information visualisation (infovis) has a great potential to allow users to explore their data on-the-go. However as the display space of mobile devices tends to be more compact, mobile infovis apps are not able to display larger amounts of information on-screen at any one time and need to depend on interaction for more in-depth exploration of the data. Despite this, existing mobile visualisation interfaces rarely include the more interactive aspects of infovis such as direct manipulation [4, 30], dynamic filtering [34, 38], and details-on-demand [34, 38].

Unlike traditional mobile interface components, which can count on established interaction patterns and gestures that are known to most people [2, 16], due to the diversity and relative novelty of mobile infovis techniques, there are no current sets of gestures that can be considered standard for carrying out infovis-specific tasks on infovis components [31]. This absence of standardised gestures and interaction patterns requires designers to carry out extensive evaluations to ensure that the gestures chosen for carrying out infovis tasks within mobile apps are intuitive, ergonomic and, overall, usable. Yet, evaluating mobile infovis interaction designs at the early stage is challenging because, on the one hand, mobile infovis needs to rely on interactions to optimise the screen usage while displaying data [37]. On the other hand, it also needs to support animations to allow the users to maintain orientation during view transitions [9] while ensuring usability in the dynamic contexts that characterise mobile device use [23]. To that end, there are no established methodologies that support the evaluation of all the above-mentioned peculiarities that characterise mobile infovis. Similarly, the support for standard gestures and animations offered by commercially available tools commonly used for prototyping (e.g. Figma, Balsamiq, PowerPoint) is mainly tailored to traditional mobile interface components (e.g. [3, 14, 20]), ultimately making these tools less practical for managing the potentially large number of interactive and animated components that compose an infovis.

To address the shortfalls of existing tools and methodologies, we are developing Sketch-A-Touch, a proof-of-concept for a tool to allow designers to generate interactive animated mock-ups for mobile infovis applications in a time-efficient, cost-effective and code-less manner. To achieve this ambitious goal, our tool combines the strengths of low-fidelity prototyping techniques with the sprite animation technique used by arcade videogames [19] and the gesture support offered by high-fidelity prototypes.

In this late-breaking work, we present the results of a formative study conducted with five experts within the fields of infovis, industrial design and videogame development on a preliminary version of the Sketch-A-Touch interface for mixed fidelity mobile infovis interaction design. Specifically, we present the challenges faced while simulating the use of our tool for generating interactive animated prototypes of two mobile infovis apps. Each of these uses different representations of hierarchical data where a variety of actions are needed and there are no established conventions for mapping gestures to actions. The results of our evaluation demonstrate the potential of mixed fidelity approaches for improving mobile infovis interaction design and how the challenges we encountered extend beyond the specific limitations of our tool.

2 RELATED WORK

Early-stage evaluation of mobile applications is widely acknowledged as a crucial aspect of app development [36], as it not only fosters the generation of design ideas but also empowers designers to assess various solutions and identify the one that aligns best with the specified requirements and user expectations. Low-fidelity prototypes are a good approach for this purpose, as they allow designers and developers to quickly generate numerous prototypes while investing a relatively modest number of resources [35, 36], as opposed to hi-fi prototypes [37]. Techniques such as paper prototyping and wireframing are two examples of lo-fi prototyping techniques that are widely used by both research and industry [35] thanks to their being quick and cheap to produce and accessible to anyone, as they just require pen and paper. Furthermore, their simplicity and sketch-like appearance proved to make users feel more engaged in the design process and comfortable providing constructive feedback on potential design improvements [7, 25]. This stands in contrast to more polished and finalised hi-fi prototypes, whose look might discourage extensive critique from users due to the perception that significant changes may be less feasible [25].

Despite the advantages, when applied to mobile app design these lo-fi prototyping techniques are less effective [11, 13]. In fact, the physical form and mobility of a mobile device are integral to its usability and overall functionality. This makes it difficult for the interface, specifically its interaction [25–27, 33], to be simulated without using the device itself. Simulating real-world conditions, however, means that the prototype must offer an automated and fluid interaction [25], ideally without requiring a Wizard of Oz to manually swap the views. Despite this challenge has been addressed by works like [5, 8, 12], the proposed approaches are still not appropriate for prototyping mobile infovis apps as they offer very limited support for touch gestures and animations. These two aspects are very important for mobile infovis because, unlike traditional mobile apps, visualisation apps are made of a large number of (possibly interactive) visual elements that need to be animated in order to maintain the user-oriented during interaction and view transitions [37, 38]. Yet, if studies on mobile infovis exploring the advantages and challenges associated with gesture interaction are many (e.g. [4, 17, 18, 22, 30–32]), most of them rarely, if ever, mentions or discusses in depth the methodology behind gesture-task selection and whether early prototypes have been used.

Considering the tools available on the market, designers are presented with a wide choice that spans from tools specifically designed for mobile app prototyping (e.g. Figma, Sketch, AdobeXD) to tools originally designed to serve different functions, which are repurposed for prototype development (e.g. Microsoft PowerPoint [21, 24, 26]). With these tools, designers can create prototypes of mobile applications with different degrees of interactivity (limited for PowerPoint, which only supports tap/click, advanced for tools like Figma, which also supports standard gestures like long-press, swipe and drag) and more or less advanced animated transitions. The majority of these tools, however, is specifically tailored to support the interaction with- and the animation of traditional mobile app components, whose visual properties and layouts follow specific standards (e.g. [1, 15]). This stands in contrast to infovis components, whose properties, layouts, and animations are mainly data-driven [37], thus more difficult to standardise. Simulating complex animations with the above-mentioned tools would require the designer to manually organise the visual elements that compose the visualisation into layers and hierarchies, and then animate each item individually. While this activity could be considered more accessible than coding high fidelity interactive animated prototypes [33], especially for designers with basic or no coding skills, it would still be time consuming to carry out [37], ultimately jeopardising the benefits of early-stage prototyping.

3 SKETCH-A-TOUCH

The motivation behind the development of Sketch-A-Touch stems from some challenges we faced while designing the interaction for two infovis apps for smartphone devices, namely a visual file browser and an activity tracker app. In order to optimise the screen usage, we opted for implementing direct manipulation of the visual elements composing the visualisations by means of standard gestures, as recommended by previous literature (e.g. [4, 29, 32]). However, during the process of deciding which gesture to use for carrying out infovis tasks [38] such as Explore or Zoom-in/Zoom-out, we found that multiple combinations of standard gestures could have been a potentially a good fit. These gestures in fact satisfy both the established guidelines for (traditional) mobile app design (e.g. [2, 16]), and the recommendations of previous literature concerning discoverability and consistency [4, 22, 29, 31, 32]. Evaluating all these possible designs, ideally with a wide audience, was thus crucial to ensure the ergonomics and intuitiveness of our visualisation apps.

From a discussion with two infovis experts on the best ways to evaluate our designs, we identified four requirements to use for searching a tool or a technique, among the existing ones, that allowed us to generate interactive animated prototypes. We considered approaches ranging from paper prototypes, to tools like Adobe Illustrator and Figma, to infovis libraries like D3.js, yet none of them comprehensively addressed all our requirements. We thus opted to transform our four requirements into the design goals listed below to design a tool specifically tailored to the creation of mobile infovis prototypes.

DG1 Support rapid low-fidelity prototyping. Designers should be able to quickly transform early concepts into tangible, low-fidelity prototypes in order to gather feedback from both

the users and other design team members (e.g. developers [24, 37]) in a time-efficient and cost-effective manner.

DG2 Support standard gestures and animations. Gestures and animations allow designers to optimise the screen usage while maintaining a high degree of usability despite the challenges posed by the mobile context, i.e. the limited attention [23, 28, 33], varied ergonomics [25, 28] and dynamic environmental conditions [26–28].

DG3 Support code-less prototyping for mobile device deployment. Designers, including the ones with basic or no coding skills, should be able to generate interactive animated prototypes that can be run directly on mobile devices in order to simulate real-world conditions [26].

DG4 Support remote evaluations with asynchronous methods. Designers should be able to gather feedback from a potentially global audience as a complement to that obtained from traditional face-to-face evaluations, so to ensure that gestures are discoverable, intuitive and ergonomic across different cultures [28].

3.1 Framework

From the design goals listed above we derived the Sketch-A-Touch proof-of-concept, which has been implemented in Typescript as an Angular web-app and can be run on a local machine. The idea behind this tool is to allow designers to sketch their visualisations on paper, in a storyboard-like fashion, set up the interaction and the animations through a graphic interface (Fig. 1) and obtain a source code for deploying a mixed-fidelity prototype that can be

run directly on the target devices, by leveraging commercially available web-page deployment tools like CodePen. This process enables designers, regardless of their coding proficiency, to evaluate interaction designs directly on real devices, potentially reaching a global audience also thanks to the web-tailored source code, which can be embedded into existing online questionnaires to create custom interactive questions.

4 FORMATIVE STUDY

In order to evaluate our concept of using mixed fidelity prototyping to improve the process of developing interactive mobile visualisation interfaces, and better understand how this could be realised, we conducted a formative study with our Sketch-A-Touch prototype using an exploratory testing approach. This involved five experts within infovis (3, IV-E), industrial design (1, ID-E) and video game development (1, VD-E) fields to interact with a preliminary version of Sketch-A-Touch using both pre-drawn sketches, derived from the infovis apps mentioned in section 3, and allowing the experts the flexibility to create and test their own cases. For each sketch, the experts followed the process illustrated in Fig. 1 to generate interactive animated prototypes and test the concept of mixed fidelity prototyping. Throughout the evaluation, we engaged in open dialogue, fostering an environment where experts freely shared their thoughts and engaged in focused discussions when specific issues or challenges arose.

In its current version, Sketch-A-Touch offers only a limited ability to automatically generate working code for certain gestures or complex combination of gestures. Whenever feasible, we tried to

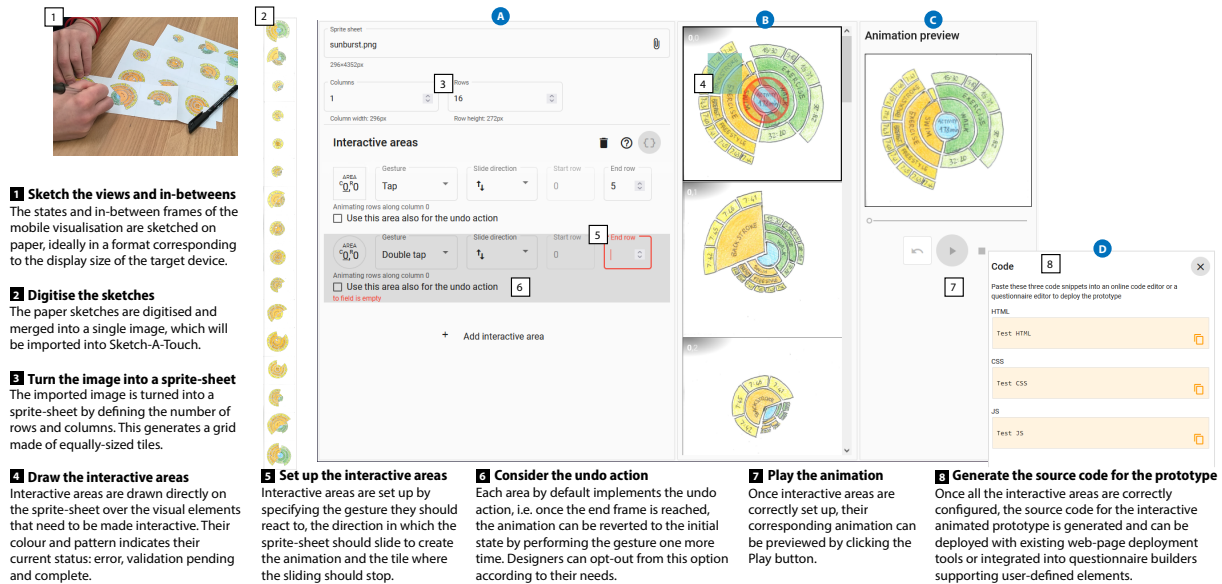


Figure 1: The interface of Sketch-A-Touch is organised as a pipeline (steps 3–8, specifically) to guide the designer in turning their sketches into a sprite-sheet and in setting up the interactions and animations that Sketch-A-Touch will then translate into HTML/CSS/JS code. The interface consists of three main blocks: (A) control panel, (B) sprite-sheet view, and (C) animation preview panel, and a drawer (D) to display the source code for the prototype. To minimise potential mistakes by the designer and the deriving delays, the sections of each block, as well as the drawer, are enabled gradually as the designer successfully completes all the previous sections (DG1)

manually code the interaction directly on the prototypes generated by the experts to foster a comprehensive discussion on all the aspects of the prototyping process and even on the future use of these prototypes (e.g. within in-person evaluations or even remote evaluations by means of online questionnaires).

Our formative study revealed a number of challenges that seem to extend beyond the specific limitations of our tool.

4.1 Test Case 1: Activity Tracking App

The first application used as test case for our exploratory test is an activity tracker displaying data about the number of steps and the duration of exercise sessions. The visualisations for this app were chosen based on a broader ongoing study on user preferences about interactive infovis for smartphones, and feature a histogram, a calendar heatmap and a sunburst.

4.1.1 Histogram and Calendar Heatmap. The number of steps made each day is displayed as a histogram and as a calendar heatmap (Fig. 2). Both visualisations feature the zoom functionality, enabling users to examine cumulative step counts such as the total steps taken each week or on Mondays of each month. The objective of these two charts was to assess the intuitiveness and ergonomics of contextual gestures [31]. As such, we needed to generate multiple prototypes, each incorporating the same gesture (e.g. pinch), but applied to different areas of the visualisation.

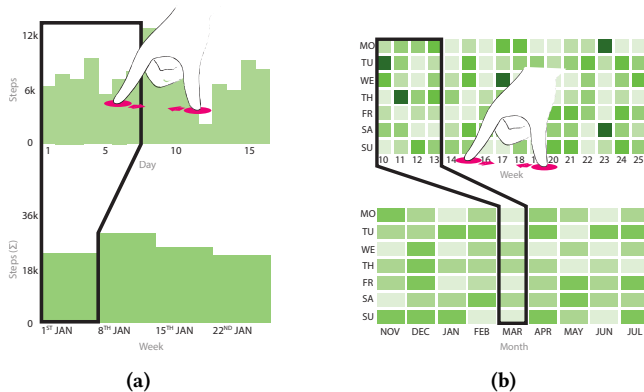


Figure 2: The histogram and the heatmap leverage the pinch gesture to reconfigure the x-axis to display aggregated data. The goal of the prototypes to be generated was that of testing the intuitiveness of contextual gestures. This example illustrates two possible contexts for pinch. In (a) the interactive area reacting to the pinch gesture is set on the x-axis, whereas in (b) it is positioned directly on the heatmap.

This test case highlighted the challenge (C1) of a mismatch between different gesture implementations in different application environments. It showed us how different environments detect certain gestures, such as pinch, but also swipe and drag – some environments can record details such as the distance each digit has travelled, the speed, the pressure, etc., whereas others can only detect gesture activation at a specific screen location. This is problematic as an interface could be designed and evaluated for

actions that cannot be properly implemented in the target application environment, or even the gestures that work best for the target application cannot be implemented in the prototype.

This test case had the benefit of allowing us to observe how our experts, acting as test users, followed different approaches in selecting the contexts and the gestures for this test case, with those having basic (e.g. IV-E3) or no coding skills (e.g. ID-E) being more creative than the experts with more advanced coding skills (e.g. VD-E). In fact, the latter seemed to instinctively restrict the choice of gestures or interactive areas to few options, likely due to their heightened awareness of potential unexpected behaviours arising from inconsistent gesture detection across different environments. These different approaches suggest that designers need to have a good understanding of how the gestures work in different environments or better communication with developers who have this knowledge (as recommended by Walny et al. [37]). This should form part of a more cohesive approach to the application design that accounts for aspects such as the application environment, user preferences, and the context in which the application is to be used. It also indicates that a more consistent implementation of gestures within different environments is needed, or that gesture detection in different application environments should be better communicated so that the potential functionality of different gestures can be better understood.

4.1.2 Sunburst. The duration of daily exercise sessions are displayed in a sunburst, with sectors indicating the activity type (e.g. swim, walk), specific exercise details (e.g. breaststroke, backstroke), and session duration. By interacting with the sectors and with the centre of the sunburst, the user can navigate the structure and reveal details of the descendants of each sector (such as the overall duration of all of a particular day's backstroke sessions).

The interaction design process for this visualisation revealed a general challenge with the potential for certain gestures to be incompatible when used together on different areas of the interface (C2). Certain combinations of gestures in overlapping areas or gestures activating on areas too close to each other could cause problems for the interface to detect the gestures, or for the user to make accurate selections. In our test case, the backward navigation of the hierarchy could be implemented by interacting with the centre of the sunburst (see Fig. 3) with different gestures to navigate either up just one level (B) or to the root of the hierarchy (A). For instance, using a tap and double tap respectively, the action for the tap could be triggered at the start of a double-tap action. Similarly, if the choice for this pair of gestures included a pinch, pinching the sunburst centre could potentially trigger actions associated to the sectors of the first ring. While these challenges might seem trivial to solve, for example by selecting more easily distinguishable gestures or implementing a new feature in Sketch-A-Touch to support areas that can respond to multiple gestures, these solutions may well decrease the tool's flexibility or introduce too much complexity for designers, thus undermining DG1 and the benefits of low fidelity prototyping.

These types of conflict can be subtle to spot or predict, especially for designers who lack sufficient experience with touch devices or coding interactions. If ignored or mishandled (for instance, expert

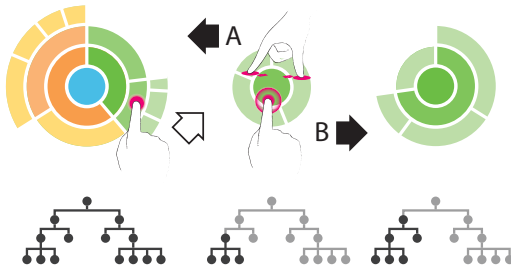


Figure 3: The sunburst allows the user to unveil/conceal details about the duration of exercise sessions based on activity and exercise type. This image illustrates an example of prototype featuring a tap gesture on a sector to navigate to the details for that session. Interacting with the sunburst centre prompts the question “Where should we navigate to?”. Here pinch-in allows us to navigate back to the root (A) while with double-tap we navigate back to the parent of the current node (B). The overlapping between the two interactive areas or even with areas of nearby sectors might decrease the usability of the resulting prototype.

ID-E tried to overcome overlaps by reducing the size of interactive areas), overlaps and conflicts could considerably decrease the prototype usability and exacerbate other usability issues (in this case, the fat-finger problem [4, 9, 31] and inaccurate selection due to dynamic context [9]). Ultimately, designers need to be aware of potential conflicts and overlaps when determining the actions to implement for making a visualisation interactive, and they also should test these actions as early as possible with target users and ideally in real-world conditions for better results [26]. If the designers are not intuitively aware of these potential conflicts through their own experience and ability, then prototyping tools could help them with this, for example by displaying tips or warnings whenever appropriate.

4.2 Test Case 2: Visual File Browser

The second app used for our test cases is a file browser based on previous works [6, 10] on fish-eye distortion for implementing overview+details visualisations for smartphone devices. The visualisation is a space-filling tree displaying the directory structure of the smartphone’s file system that can be navigated by directly touching the nodes or by moving the fish-eye lens over nodes of interest (Fig. 4). Both these interactions act on the fish-eye distortion lens, making target nodes push their siblings towards the edges of the screen to free up space for their descendants. Testing interactions for this behaviour revealed two distinct challenges for our proposed prototyping approach.

To obtain a faithful representation of the fish-eye distortion behaviour, the underlying hierarchy of data has to be composed by numerous nodes. Yet, drawing a large number of nodes not only requires time but also precision (challenge C3). To investigate this issue we tried different strategies using sketches with different levels of detail. The animation for the less precise drawings was poor as the node expansion was hardly noticeable. As indicated by Walny et al. [37], the scale and complexity of the data determines the

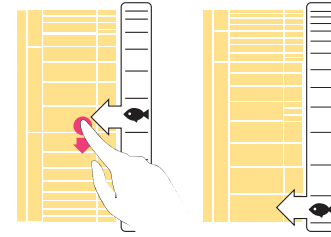


Figure 4: The space-filling tree leverages the fish-eye distortion to optimise the screen usage and implement the overview+details paradigm [6, 9]. A drag gesture would be needed to move the fish-eye distortion lens over the nodes to allow exploration of other portions of the tree. However, animating the view as the finger touches the screen remains an open challenge because directly mapping the finger’s position to the sprite-sheet animation could yield sub-optimal results.

complexity of the visualisation and this should be accommodated even at the prototyping stage. Drawing precise in-between frames in these cases could result in a daunting amount of work, breaking the ethos of rapid prototyping (DG1). Our second infovis expert (IV-E2) suggested to use generative AI tools to draw the in-between frames. We tested this suggestion with Bing’s Copilot (powered by Dall-E 3) feeding it the initial and final frames of a simple chart (the histogram of Test Case 1) with a request for it to draw 3 in-between frames. The result was somewhat creative, but could not be considered precise (see supplemental material) indicating that common publicly accessible AI tools may not yet be ready for this sort of task.

The reflection we drew from this challenge is that sketching visualisations can be an intrinsically complex and time-consuming process, regardless of the designer’s ability or the approach used to draw the frames. While existing tools and libraries, such as D3.js, Tableau, Illustrator, or Photoshop, allow designers to achieve a trade-off between data-realistic visualisations and the time required for frame creation, novel or custom visualisations incorporating uncommon visual elements or animations (such as our fish-eye distortion effect) might still face limitations in leveraging these resources. Yet, the benefit of interaction design tools like Sketch-A-Touch, Figma or PowerPoint is that they allow designers to explore different methods to produce the animations (such as creating the frames in stop motion by manually rearranging physical paper prototype components, or different generative AI implementations).

An additional challenge we faced during the formative study is that the actions associated to certain gestures, specifically those involving the continuous movement of fingers on the screen, cannot be faithfully simulated using sprite-sheet animations (C4). In our test case, the effect of the fish-eye distortion depends on gesture attributes (e.g. start point, end point, speed and direction) that are hard to predict when sketching the frames. Attempting to represent these interactions with a fixed number of frames would likely lead to a discrepancy between the user’s gesture and the smoothness of the animation, making the prototype less effective for evaluations (DG4). This potential mismatch between gesture and action would

surface for any gesture than does not have a predictable effect, such as scrolling or brushing, where the end point of the interaction and the item(s) the user will select cannot be determined in advance.

To address this challenge, designers could guide users by somehow telling them what to do, e.g. “*drag up to scroll*” or “*select item X*”. However, for more complex gestures (such as dragging to select a group of items) it might not be so easy for designers to communicate the desired gesture, or for the user to realise it. For these complex or hard-to-describe gestures, the prototypes created with Sketch-A-Touch are likely to be less effective, requiring the user to use more of their imagination to envision how the gesture might be implemented in the final version of the infovis app. While prototyping tools such as Figma allow these sorts of actions to be tested to an extent, a considerable additional effort is required to the designer for configuring the test. Indeed, one of our experts, VD-E, described how continuous gestures like scrolling or dragging could be implemented in Sketch-A-Touch using different layers to obtain a sort of parallax scrolling. However, this solution would as well involve additional steps to configure the test. Ultimately, there would be a trade-off between the degree to which gestures are simulated and how much effort the interaction designers put into configurations, requiring a cost-benefit evaluation to assess the feasibility of this solution.

5 CONCLUSION AND FUTURE WORK

In this late-breaking work we introduced Sketch-A-Touch, a proof-of-concept for a tool to enable mobile information visualisation designers to easily generate interactive animated prototypes to evaluate mobile infovis interaction design. We presented the challenges that emerged during an exploratory study conducted on a preliminary version of our tool and the reflections we drew from them. While these results will help us improve Sketch-A-Touch, they also demonstrate how the challenges we encountered are intrinsic to mobile information visualisation. We believe these results can serve as catalysts for further discussions, investigations and improvements in the field of mobile infovis prototyping.

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